

Scenario-Based DevOps Interview Cheat Sheet

Real-world troubleshooting & on-call scenarios exactly like actual interviews.

1. Kubernetes Scenarios

Scenario 1: "Your pod is in CrashLoopBackOff. What do you do?"

Answer Path:

1. Check pod events

```
kubectl describe pod <pod>
```

2. Check logs

```
kubectl logs <pod> --previous
```

3. Verify:
 4. Probes failing?
 5. ConfigMap/Secret wrong?
 6. Image tag incorrect?
 7. Resource limits too low?
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Scenario 2: "Pods stuck in Pending."

Answer Path:

1. Check node capacity (CPU/RAM)
 2. Check taints/tolerations
 3. Storage class missing
 4. Pod affinity blocking scheduling
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Scenario 3: "Service not reachable inside cluster."

Answer Path:

1. Check service selector
2. Check endpoints

```
kubectl get endpoints <svc>
```

3. Check network policies
4. DNS resolution

```
kubect1 exec -it <pod> -- nslookup service
```

2. Docker Scenarios

Scenario 1: "Container keeps restarting."

Answer Path:

1. Check container logs
2. Check healthcheck failures
3. Validate environment variables
4. Verify entrypoint/CMD
5. Check port conflicts

Scenario 2: "Image works locally but fails in container."

Answer Path:

- App listening on 0.0.0.0?
- Missing dependencies?
- Wrong file paths?
- ENV variables missing?
- Case sensitivity (Linux FS)?

3. AWS Scenarios

Scenario 1: "EC2 suddenly not reachable."

Answer Path:

1. Check SG inbound rules
2. Check NACL outbound/inbound
3. Check EC2 system status checks
4. Use SSM Session Manager
5. Check disk full or CPU 100%

Scenario 2: "S3 Access Denied."

Answer Path:

- Check IAM
 - Check bucket policy
 - Block Public Access enabled?
 - Wrong region?
 - KMS encryption issues?
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Scenario 3: "RDS connection failing."

Answer Path:

1. Security group DB → EC2
 2. PublicAccessible=True?
 3. Test port connectivity
 4. Check credentials & rotation
 5. Check max_connections limit
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4. Terraform Scenarios

Scenario 1: "Resource deleted manually in cloud."

Answer Path:

- Terraform will detect drift
 - Run `terraform apply` to recreate
 - Check CloudTrail who deleted
 - Enforce IaC-only rules
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Scenario 2: "State file corrupted or missing."

Answer Path:

1. Restore from S3 versioning (if remote)
 2. Use `terraform.tfstate.backup`
 3. Import resources manually
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5. Jenkins Scenarios

Scenario 1: "Pipeline stuck at a stage."

Answer Path:

- Check agent logs
 - Workspace permission issues
 - Missing tool installation
 - SCM webhook loop
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Scenario 2: "Job not triggering after Git push."

Answer Path:

- Webhook delivery errors
 - Jenkins URL unreachable
 - CSRF protection settings
 - Branch filters incorrect
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6. Linux Scenarios

Scenario 1: "Server running slow."

Answer Path:

1. Check CPU:

```
top / htop
```

2. Check disk:

```
df -h
```

3. Check logs:

```
journalctl -xe
```

4. Check I/O wait

Scenario 2: "Port 80 not responding."

Answer Path:

1. Check service running
2. Check firewall/iptables
3. Verify process listening

```
ss -tulnp | grep 80
```

4. Check reverse proxy config
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7. Networking Scenarios

Scenario 1: "DNS not resolving."

Answer Path:

- Check /etc/resolv.conf
- Test:

```
dig google.com  
nslookup google.com
```

- Check DNS server availability
-

Scenario 2: "Website reachable internally but not externally."

Answer Path:

- Security groups / firewall
 - Reverse proxy
 - NAT gateway issue
 - Routing table misconfiguration
-

8. DevSecOps Scenarios

Scenario: "Docker image has vulnerabilities."

Answer Path:

1. Scan with Trivy
2. Upgrade base image
3. Remove unused packages

4. Implement multi-stage builds